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The utility of microspheres as surrogates for the transport of *E. coli* RS2g in partially saturated agricultural soil

Joanna M. Passmore [a,1], *David L. Rudolph* [a,*], *Maria M.F. Mesquita* [b,2], *Edwin E. Cey* [c,3], *Monica B. Emelko* [b,4]

[a] *Department of Earth and Environmental Sciences, University of Waterloo, 200 University Avenue West, Waterloo, Ontario N2L 3G1, Canada*
[b] *Department of Civil and Environmental Engineering, 200 University Avenue West, Waterloo, Ontario N2L 3G1, Canada*
[c] *Department of Geoscience, University of Calgary, 2500 University Drive NW, Calgary, Alberta T2N 1N4, Canada*

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ABSTRACT

Polystyrene latex microspheres are widely used as surrogates for biocolloid transport in porous media; however, relatively few studies directly compare microsphere transport with that of the microorganism it is intended to represent, particularly at the field scale. Here, we compared the transport behaviour of a bacterium (*Escherichia coli* RS2g; 1.2 μm in diameter) and three different sized microspheres (1.1, 3.9, and 4.8 μm in diameter) within undisturbed agricultural field soil following infiltration under partially saturated conditions. The soil contained significant macroporosity. A tension infiltrometer was used to control the application of a transport solution containing Brilliant Blue FCF dye to two plots. A >2 log reduction in the concentration of all colloids was observed from the soil surface to 5 cm depth in both plots. The concentration of colloids in the soil was generally proportional to the intensity of soil dye staining; however, both the *E. coli* RS2g bacterium and the 1.1 μm microspheres appeared to be transported deeper than the other colloids and the visible dye along root holes at the bottom of the profile in both plots. The similarities in size and zeta potential of the 1.1 μm microspheres and the *E. coli* RS2g likely contributed to that outcome. Colloid concentrations in dyed soil by depth were similar between the two plots, despite differences in soil properties and infiltration patterns. The properties of the colloids and macropore density were the most important factors affecting colloid transport. These results suggest that microspheres with size and surface properties similar to the microbe of interest are useful surrogates to trace potential pathways of transport in the subsurface.

* *Corresponding author*. Tel.: +1 519 888 4567x36778.
E-mail addresses: jpassmor@uwaterloo.ca (J.M. Passmore), drudolph@uwaterloo.ca (D.L. Rudolph), mmesquita@uwaterloo.ca (M.M.F. Mesquita), ecey@ucalgary.ca (E.E. Cey), mbemelko@uwaterloo.ca (M.B. Emelko).
[1] Tel.: +1 519 888 4567x37124.
[2] Tel.: +1 519 888 4567x38244.
[3] Tel.: +1 403 220 8393.
[4] Tel.: +1 519 888 4567x32208.
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1. Introduction

The passage of microorganisms through the unsaturated zone is of concern where there are potential pathways of rapid transport to ground or surface water, particularly when those water bodies are untreated or inadequately treated sources of drinking water. Between 1991 and 2000, 68 outbreaks of waterborne illness in the USA were associated with drinking water drawn from groundwater aquifers, and the major deficiency in these systems was deemed to be source water contamination (USEPA, 2006). In a Canadian survey of rural drinking water wells in Ontario, bacterial contamination was detected in more wells than contamination by pesticides or nitrate (Goss et al., 1998). One of the most widespread microbial threats to groundwater and surface water quality comes from the potentially harmful microorganisms in manure and other biosolids applied to agricultural land (Gerba and Smith, 2005).

Microbial transport and survival in the subsurface, particularly the vadose zone, are not well understood and have potentially serious implications for drinking water quality. Although fecal indicator or pathogen concentrations in source water may be low in many cases, concern still exists because some pathogens (e.g. *Cryptosporidium*, *Escherichia coli* O157) have a low human infective dose (i.e. the ingestion of only a few microorganisms may cause illness) (DuPont et al., 1995; Strachan et al., 2001). Where water tables are shallow or conditions of thin overburden over fractured rock aquifers exist, the risk of groundwater contamination by pathogenic microorganisms is particularly high. Surface water sources may also be put at risk from agricultural land amendments where tile drains transport soil water rapidly from the shallow subsurface to adjacent surface water systems (Coelho et al., 2007).

Preferential flow is generally accepted as a ubiquitous property of soils that enables the short-circuiting of natural retention mechanisms in soil; thereby allowing contaminants a quick pathway to source waters (Unc and Goss, 2003; Coelho et al., 2007; Burkhardt et al., 2008; Cey et al., 2009). The risk posed to groundwater by preferential transport through the vadose zone is poorly understood, however, and needs to be further characterized. The use of latex microspheres in controlled tracer tests has proven to be a useful tool for investigating these phenomena. At the lab-scale, microsphere transport has been compared with that of live microorganisms in unsaturated (Sirivithayapakorn and Keller, 2003) and saturated columns filled with sand (Lindqvist and Bengtsson, 1995), granular activated carbon (Paramonova et al., 2006), and other granular media such as anthracite and garnet (Emelko and Huck, 2004). Similar experiments have been conducted using cores of undisturbed natural field soils containing macroporosity features using lysimeter systems (Passmore, 2005; Sinreich et al., 2009).

A limited number of experiments have been conducted in the field under saturated conditions in fractured rock (McKay et al., 2000; Auckenthaler et al., 2002; Becker et al., 2003) and sandy aquifer sediments (Harvey et al., 1993, 1995; Bales et al., 1995). There are also few field studies in the scientific literature that have employed microspheres as surrogates for microbial transport in natural soils under partially saturated conditions (Burkhardt et al., 2008; Cey et al., 2009). Although useful conceptual information has clearly been derived from the interpretation of microsphere transport in field settings, it remains uncertain how representative these observations are with respect to the transport behaviour of live microorganisms. As a further complication, standard methods for colloid sampling and enumeration in soil are also lacking (McCarthy and McKay, 2004).

This study builds on previous work in which conditions of preferential flow in undisturbed field soil were elucidated by the application of a dye tracer under tension (Cey and Rudolph, 2009) and the associated transport of microspheres was assessed under these conditions (Cey et al., 2009). The main objectives of this work were: (1) to develop an experimental protocol for studying concurrent microbe and microsphere transport in natural soils under partially saturated conditions in the field; and (2) to compare microbe and microsphere transport characteristics in unsaturated agricultural soils to evaluate the utility of microspheres as surrogate tracers of bacteria.

2. Materials and methods

2.1. *Field site*

Infiltration experiments were conduced in an agricultural field near Kintore, Ontario, Canada (Fig. 1) in August of 2008. The soil in the field was predominantly a well-drained Honeywood silt loam (grey-brown Luvisol) consisting of silty deposits over calcareous loam till (Wicklund and Richards, 1961). No-till soybeans were cultivated the previous summer, and no-till winter wheat had been harvested 2 weeks before this study began. Two infiltration locations, designated P1 and P2, were selected based upon differences in soil texture and structure. Depth to the water table was approximately 0.90 m at P1 and 1.17 m at P2 at the time of each experiment.

2.2. *Colloid selection and characterization*

Two types of colloids were selected for use during this investigation: *E. coli* RS2g bacteria (Saini et al., 2003) and polystyrene microspheres of various size and surface characteristics. Colloid properties are listed in Table 1. *E. coli* RS2g is resistant to two antibiotics (kanamycin and rifampicin) and expresses the green fluorescing protein (GFP) gene, which results in bacteria colonies that fluoresce bright green under UV (365 nm) light. Pure cultures were maintained at $-80\,^{\circ}C$ in Luria–Bertani (LB) Lennox broth (20 g L^{-1}, Fisher) containing two antibiotics (100 $\mu g\ mL^{-1}$ kanamycin and 10 $\mu g\ mL^{-1}$ rifampicin) and 10% glycerol. To prepare *E. coli* RS2g bacteria for use in experiments, 1 mL of culture stock was grown to stationary phase in 100 mL of LB-antibiotic broth using a water bath shaker at 37 °C and 150 rpm for ~15 h, at which point the cells were centrifuged down to a pellet (for 10 min at $5000 \times g$) and washed twice in sterile 1X phosphate buffered saline (PBS; pH 7.4) before resuspension in PBS. The nominal size of the bacteria was determined using two approaches: (1)

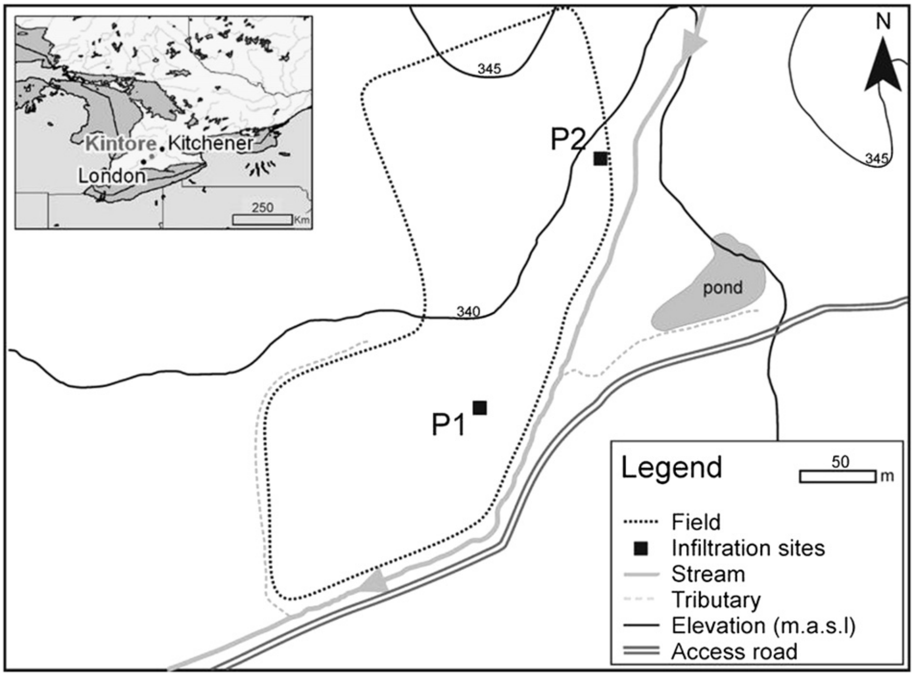

Fig. 1 – Location of the field near Kintore, Ontario, and site plan of the two infiltration experiments.

image analysis (Northern Eclipse, Mississauga, Ontario) and phase contrast microscopy to determine the average lengths of the major and minor axes of the cells; and (2) dynamic light scattering (DLS) (Zetasizer Nano ZS, Malvern) to determine the equivalent spherical diameter.

Three different polystyrene microspheres (MS) were used during this investigation. The microspheres had mean diameters of 1.10 (MS-1), 3.87 (MS-2), and 4.87 μm (MS-3) and densities of 1.05, 1.07, and 1.05 g cm^{-3}, respectively, as reported by the manufacturers. The nominal microsphere sizes were measured using DLS (Zetasizer Nano ZS, Malvern) to determine the equivalent spherical diameters. MS-1 was used because it was similar to *E. coli* RS2g in size. MS-2 and MS-3 were larger and approximate the size of protozoan oocysts of *Cryptosporidium*. Both MS-1 and MS-3 had surfaces modified by the addition of short-chain carboxylic acids, giving them a net negative charge as indicated by their zeta potential. MS-2 had an unmodified polystyrene surface, giving it a weak negative charge at the pH and ionic strength of the transport solution used.

Microelectrophoresis (ZetaSizer Nano ZS, Malvern) was used to determine the electrokinetic properties of the colloids used in this investigation. Electrophoretic motility was measured at 20 °C using colloid suspensions (4.1×10^6 *E. coli* RS2g cells mL^{-1}, 1.0×10^7 MS-1 mL^{-1}, 1.0×10^5 MS-2 mL^{-1}, and 1.0×10^5 MS-3 mL^{-1}) prepared in the electrolyte solution (0.10 M KCl). These measurements were repeated using at least three different samples of each colloid suspension. Zeta

Table 1 – Properties of colloids used in infiltration experiments.

Colloid	Surface properties	Size (μm)	Zeta potential (mV)[a]	Fluorescing colour	Excitation/emission (nm)[b]	Manufacturer
E. coli RS2g	n/a	2.5 by 0.8[c] 1.16 ± 0.02[d]	−12.1 ± 0.67	Green	365/510	n/a
MS-1	Carboxylated	1.10 ± 0.04[e] 1.63 ± 0.41[d]	−18.0 ± 2.23	Red	535/575	Invitrogen Corp., Eugene, OR
MS-2	Plain	3.87 ± 0.025[e] 3.17 ± 2.19[d]	−3.7 ± 1.43	Purple	360/420	Bangs Laboratories, Fishers, IN
MS-3	Carboxylated	4.87 ± 0.25[e] 16.03 ± 18.19[d]	−37.8 ± 2.80	Green	441/486	Polysciences Inc., Warrington, PA

a Measured at 20 °C in 0.10 M KCl, reported as the mean ± standard deviation.
b Wavelength of maximum excitation and emission.
c Mean lengths of major and minor axes of the rod-shaped cells as determined by image analysis.
d The equivalent spherical diameter as measured by dynamic light scattering (DLS) at 20 °C in 0.10 M KCl, reported as the mean ± standard deviation.
e Diameter as indicated by supplier, reported as the mean ± standard deviation.

potentials were calculated from electrophoretic mobilities using Smoluchowski's equation. Unfortunately, due to concerns regarding equipment contamination, the electrokinetic properties of the colloids could not be analyzed with the Brilliant Blue dye tracer in solution (described in Section 2.5).

2.3. Method development to recover colloids from soil

Prior to the infiltration experiments, preliminary laboratory experiments were conducted to develop a reliable method for the recovery of *E. coli* RS2g and microspheres from soil. Soil aggregates must be broken up and colloids suspended in solution using a mechanical method such as shaking, blending, or sonication. Sonication is a quick and effective way of dispersing soil aggregates and colloids, but may damage cellular membranes if exposed for too long or if the vibration is too strong. Colloid recovery from unsterilized soil from the A horizon at the field site (0–22 cm depth) was compared with recovery from unsterilized well-sorted fine sand. Three 10.0 g samples of each soil were placed in vials and 1 mL of a tracer solution containing 0.10 M KCl with 4 g L^{-1} Brilliant Blue dye and colloids was added to 10.0 g of soil. The initial concentration (C_0) of colloids in soil was 1.6×10^5 *E. coli* RS2g cells g^{-1} soil, 5.2×10^4 MS-2 g^{-1} soil and 5.0×10^4 MS-3 g^{-1} soil. For detailed consideration in this set of experiments, the two larger microspheres, representing two different surface properties, were included. The mixture was homogenized and allowed 2.5 h of contact time at room temperature. Diluent solution (20 mL of PBS and 0.01% Tween 80) was pipetted into each vial. Dispersion treatments were applied in to each vial and samples taken after each successive treatment: shaken for 20 s, sonicated for 10 s, and sonicated for an additional 20 s. Sonication (Branson Sonifier 250, Branson Ultrasonics Corporation, Danbury, CT) was carried out at 40% of total output (80 W) in 1 s pulse intervals. After each treatment two 1 mL samples were collected, one diluted into 9 mL PBS for *E. coli* RS2g enumeration, the other into 9 mL 0.02% Tween 80 for microsphere enumeration. Enumeration was conducted in duplicate. The dispersion treatment with the highest recovery of RS2g cells was assessed for the recovery of microspheres, and that method used for the field experiment.

2.4. Enumeration of colloids and dye colorimetry

E. coli RS2g bacteria were enumerated using a spread plating technique during which 0.1 mL of each serial dilution was spread plated in duplicate onto LB agar containing 100 μg mL^{-1} kanamycin and 10 μg mL^{-1} rifampicin. Plates were incubated for 24 h at 37 °C prior to enumeration. Microspheres extracted from soil were enumerated by epifluorescence microscopy (Axiophot or Axiophot 2 plus, Carl Zeiss, Toronto, Ontario). A volume (1.0 or 0.10 mL) collected from the dilution in 0.02% Tween 80 was filtered in duplicate onto 25 mm diameter, 0.45 μm mesh size, gridded mixed cellulose ester membrane filters (Millipore). A supporting filter (8.0 μm, mixed cellulose ester, Millipore) was placed underneath each 0.45 μm membrane to protect it and evenly distribute the vacuum. Once the liquid was evacuated the filter was mounted onto a microscope slide and a cover slip was secured on top. Enumerations were made using the appropriate light filter for each type of fluorescing microsphere (filter sets 02, 09, and 15, Carl Zeiss, Toronto, Ontario).

The sample size (2 g of soil) was too small to conduct a separate analysis of dye intensity, so a visual colorimetric comparison was performed to classify the dye concentration in soil. After colloid enumeration the solid material was settled out of suspension and the supernatant visually classified into three categories using 12 dilutions of Brilliant Blue dye, ranging from 4 to 0.0001 g L^{-1}. The amount of dye in the original sample was calculated based on the mass of soil and volume of diluent added. The three categories were (0 – no visible dye) <0.00005 mg dye g^{-1} soil, (1 – light dye staining) ≥0.00005 and <0.05 mg dye g^{-1} soil, and (2 – heavy dye staining) ≥0.05 mg dye g^{-1} soil.

2.5. Field experiment

A tension infiltrometer (TI) (Soil Measurement Systems, Tuscan, USA) with an infiltration disc 0.209 m in diameter was used to apply the transport solution at the soil surface in a controlled manner to a defined area, based on the approach described in Cey and Rudolph (2009). The application was done under negative pressures (or tensions), requiring the soil to draw the solution from the disc via capillary forces. Time domain reflectometry (TDR) wave guides 0.100 m long were inserted into the soil along the perimeter of the disc at a 45° angle (pointing towards the center) to monitor changes in volumetric soil moisture content. A datalogger (Campbell Scientific CR10X) was used to automatically record soil water content from the TDR probes and infiltration rates from the TI tower as measured by a differential pressure transducer (Honeywell 26PC series).

Before TI installation, a small plot (0.4 by 0.5 m) was carefully cleared of surface debris and levelled if necessary, while attempting to minimize disruption of the soil structure. The 0.21 m diameter retaining ring was placed onto the ground and the colloid suspension in PBS (prepared that morning) was applied evenly over the soil surface within the ring using a dropper. The total number of *E. coli* RS2g applied was 3.7×10^{10} CFU to P1 and 3.8×10^{10} CFU to P2. The total number of microspheres applied to each of the plots was 1.1×10^8 of MS-1, 1.0×10^8 of MS-2 and 1.0×10^8 of MS-3. More RS2g cells were applied in anticipation of losses such as die-off and predation, to which microspheres are not susceptible. Fine glass beads (Spheriglass No. 2024) were applied in a layer 0.003 m deep within the retaining ring to promote good hydraulic contact between the TI disk and the soil surface (Perroux and White, 1988).

Two negative pressure potentials were applied to the TI disc, the first to wet the beads and the soil surface (P1, –20.4 cm, P2, –18.5 cm), and the second to induce flow under nearly-saturated conditions (−0.8 cm for both plots). Approximately 1 L of transport solution was applied per infiltration. The transport solution consisted of 0.10 M KCl and 4 g L^{-1} Brilliant Blue FCF dye (C.I. Acid Blue #9, 42090). The ionic strength of the transport solution approximated that of liquid manure (Unc and Goss, 2004). Characterized by Flury and Fluhler (1995), Brilliant Blue has been used in previous studies to visualize the movement of water through soil (e.g. Natsch et al., 1996; Cey and Rudolph, 2009; Cey et al., 2009). Some retardation of the dye relative to conservative ionic tracers has been reported, demonstrating some adsorption to soil surfaces (Flury and Fluhler, 1995). Here, use of the dye was critical for staining transport pathways and visualizing infiltration patterns, thereby indicating where soil samples should be collected for colloid enumeration.

2.6. Excavation, photographs, and soil sampling

Each infiltration site was manually excavated on the day following colloid infiltration and soil samples were collected for colloid analyses. The initial excavation was made along a vertical plane through the center of the infiltration site; it was photographed, and sampled. In the remaining half of the infiltration site, parallel horizontal planes were successively exposed, photographed, and sampled at depths of 0, 5, 10 cm, and every 10 cm thereafter until no more dye was visible. Excavated planes have been identified according to orientation (V-vertical, H-horizontal) and the distance (cm) from the soil surface. Each plane was carefully excavated by hand to minimize smearing and possible cross-contamination. Spatulas used to expose and smooth each plane were washed between cuts in soapy water and rinsed. A white tarp covered the excavation to reduce colloidal exposure to solar radiation and provided diffuse lighting for photography. A grey frame with 0.100 m gradations and photographic colour and grey scales were pinned to the soil. Photographs were taken with a 7.1 megapixel camera following the procedure employed by Cey and Rudolph (2009). The digital images underwent image analysis and dye intensity classification in the same manner as Cey and Rudolph (2009). The high-resolution photographs of the horizontal planes were also used to enumerate the number of macropores >2 mm in diameter. Counts were made using the entire photographed area.

Soil samples (2–5 g) were collected for colloid analysis from each exposed plane and the soil surface. A small metal spatula was washed, rinsed and flame sterilized between each sample. Samples were collected on the basis of the visible dye staining and locations were selected to represent a variety of depths, intensities of dye staining, and the presence or absence of macropores. Negative control samples were also obtained periodically from areas away from the infiltration area with no visible dye staining. Soil samples were stored on ice in the field, at 4 °C over night, and analysed the following day.

Based on the laboratory-scale method development work described above in Section 2.3, soil samples were homogenized and 2.0 g was placed into a vial containing 4 mL of a PBS and 0.01% Tween 80 solution. Each vial was shaken for 20 s and subsequently sonicated for 10 s. Serial dilutions were performed in PBS for *E. coli* RS2g and 0.02% Tween 80 for the microspheres. Limits of detection based on our methods were 10 CFU g^{-1} soil for RS2g and 10 MS g^{-1} of soil for the microspheres, and enumeration was possible over five orders of magnitude.

2.7. Survival of *E. coli* RS2g in soil

The design of the field experiment necessitated leaving the colloids in the soil for 1 day following infiltration. Sample analyses were completed 24 h thereafter. To evaluate *E. coli* RS2g survival in the soil matrices, a laboratory-scale batch experiment was performed in triplicate using unsterilized soil collected from the experimental site. Moist soil (silt loam) (10.0 g) was inoculated with cells (1.8×10^4 CFU g^{-1} soil) and 1 mL of transport solution (4 g L^{-1} Brilliant Blue dye and 0.10 M KCl). Concurrently, to investigate the effect of Brilliant Blue dye on *E. coli* RS2g survival, RS2g cells were added to a 10 mL solution of PBS and dye (4 g L^{-1} dye; 1.8×10^4 CFU mL^{-1} in suspension). Samples were homogenized and kept at room temperature. After 48 h, 20 mL of diluting solution (PBS and 0.01% Tween 80) was added to each vial. Samples were analysed and *E. coli* RS2g enumerated as described in Section 2.4.

3. Results and discussion

3.1. Laboratory experiments

Preliminary experiments were conducted to determine the best way to maximize extraction of the microspheres and *E. coli* RS2g bacteria from the soil matrices while minimizing potential inactivation of the *E. coli* RS2g by utilization of disaggregation techniques such as sonication. Fig. 2 shows the recovery of *E. coli* RS2g from the sand and silt loam using three successive disaggregation treatments. Shaking and shaking followed by 10 s of sonication yielded similar bacterial recoveries in each of the soil matrices. Sonication for an additional 20 s yielded only slight decreases in recovery. It should be noted, however, that the highest and lowest *E. coli* RS2g concentrations observed across all treatments were within one order of magnitude of each other (~0.6 log range), as opposed to the order of magnitude differences that are typically required to statistically demonstrate significant differences in microbial enumeration data (APHA, AWWA, and WEF, 2005).

Microsphere recovery was assessed using the 10 s sonication treatment. In contrast to the ~50–75% observed recovery

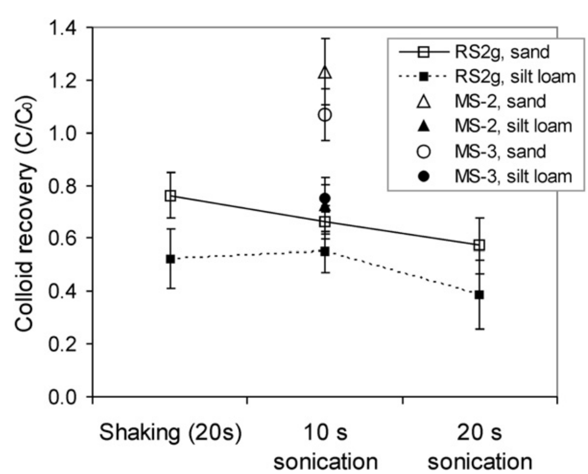

Fig. 2 – Progressive recovery of *E. coli* RS2g from silt loam and fine sand as different methods of disaggregation are applied, and recovery of microspheres MS-2 and MS-3 after 10 s of sonication from silt loam and fine sand. Presented as mean C/C_0; error bars are one standard deviation.

of *E. coli* RS2g bacteria, effectively all MS-2 and MS-3 microspheres were recovered from the fine sand, whereas recovery from silt loam ranged from ~60 to 85%. It is difficult to interpret colloid recovery data without extensive study of losses and errors associated with each method. For example, in the present study there appears to have been a general trend indicating less colloid recovery from the silt loam relative to the fine sand. This may have been observed because the silt loam contained a greater concentration of highly charged surfaces (including clay and organic matter) relative to the fine sand, and resulted in a stronger electrostatic attraction between the solid medium and the added colloids. Moreover, the smaller-sized particles of the silt loam provided more surface area for colloid retention. Because conventional statistical analysis techniques based on the *t*-distribution are often inappropriate when enumerating colloids, particularly colloids of a biological nature, more extensive recovery analysis (beyond the scope of this study) would be required to demonstrate if such potential trends are statistically significant (Emelko et al., 2008). Nonetheless, the colloid recovery results presented herein illustrate the importance of selecting an appropriate colloid recovery method for the solid medium. Based on these findings, soil samples collected in the field as part of this set of experiments were first shaken and then sonicated for 10 s to maximize the recovery of *E. coli* RS2g.

Survival of *E. coli* RS2g was explored in relation to the dye tracer and time spent in the soil environment using 48-h batch tests. Brilliant Blue dye did not have a significant effect on *E. coli* RS2g survival at the concentration used. Using the same analysis method as used during the field investigation, recovery of *E. coli* RS2g from the soil matrix was approximately 50–65% immediately after addition to the transport solution. After 2 days, *E. coli* RS2g concentrations were 40–60% of the original concentrations (essentially within the variability of the recovery data), suggesting that the variability in the *E. coli* RS2g concentrations could be predominantly attributed to enumeration methodology and that little or no inactivation of the *E. coli* RS2g occurred in the study matrices during the experimental period. The relative reproducibility achieved during the present investigation is as good as can be expected for microbial enumeration data obtained using culturing techniques from which order of magnitude differences are typically required to statistically demonstrate significant differences (APHA, ASTM, and WEF, 2005). The lack of microbial inactivation observed herein is consistent with the results of Saini et al. (2003), who studied *E. coli* RS2g inactivation in a loamy soil spiked with manure and found no change in bacterial concentrations during the first 8 days of study.

3.2. Field soil characteristics

Soil properties at each infiltration site are listed in Table 2. Although the two locations were within the same field and <200 m apart, their physical soil properties were quite different. The soil at P1 was finer-textured than at P2 and had superior soil structure development, as indicated by the greater porosity values. The higher clay mineral and organic matter content in the P1 soil resulted in a higher component of negatively charged particles, as demonstrated by the higher cation exchange capacities (CEC) in Table 2. Clay and organic matter are generally considered to contribute to the stability of macropore features in the near-surface environment, although other factors also contribute to their formation and stabilization (Horn et al., 1994). The study plots reflected this, as the number of cylindrical macropores >2 mm in diameter enumerated at P1 was almost twice that of P2 at 0, 5 and 10 cm depths (Fig. 3). The macropore density values obtained were within the range of those enumerated at other locations in Ontario (Cey et al., 2009). The enumerated macropore channels were typically earthworm burrows, which tend to have low tortuosity and high continuity to depth, and are important pathways for preferential flow and transport to depth (Jarvis, 2007).

3.3. Changes in soil moisture during infiltration

Results for each infiltration, including the applied pressure heads and mean infiltration rates, are listed in Table 3. Cumulative infiltration curves and the associated changes in volumetric soil water content for both infiltrations are presented in Fig. 4. The mean initial volumetric soil water content was very similar for both experiments (P1 = 0.22 m^3 m^{-3},

Table 2 – Soil properties by depth for each location, determined by bulk and core samples.

ID	Depth (cm)	pH[a]	OM (%)[b]	CEC (cmol$^+$ kg^{-1})[c]	Gravel (% w/w)[d]	Sand[e] (% w/w)	Silt (% w/w)	Clay (% w/w)	Texture	Bulk Density (g cm^{-3})[f]	Porosity
P1	0–22	7.0	6.5	28.4	0.5	22.9	54.6	22.6	Silt loam	1.25 ± 0.04	0.53 ± 0.01
	22–40	7.4	1.1	38.7	0.2	15.2	59.8	25	Silt loam	1.35 ± 0.20	0.48 ± 0.08
	40–55	8.0	0.5	21.6	13.2	44.2	44.1	11.7	Loam	1.76 ± 0.13	0.34 ± 0.05
P2	0–22	7.6	2.9	12.2	1.4	48.3	38.5	13.2	Loam	1.60 ± 0.06	0.40 ± 0.02
	22–40	7.9	0.7	14.5	2.5	52.3	33.4	14.3	Sandy loam	1.87 ± 0.18	0.29 ± 0.07
	40–55	8.4	0.1	10.5	10.3	51.9	41.6	6.5	Sandy loam	2.09 ± 0.06	0.21 ± 0.02

a Measured using the saturated paste method.
b Organic matter, using the Walkley–Black method.
c Cation exchange capacity, using the Barium chloride method.
d % gravel of total mass of soil, by sieving.
e % of fine fraction only, using pipette method.
f Mean bulk density ± standard deviation; measured using undisturbed samples.

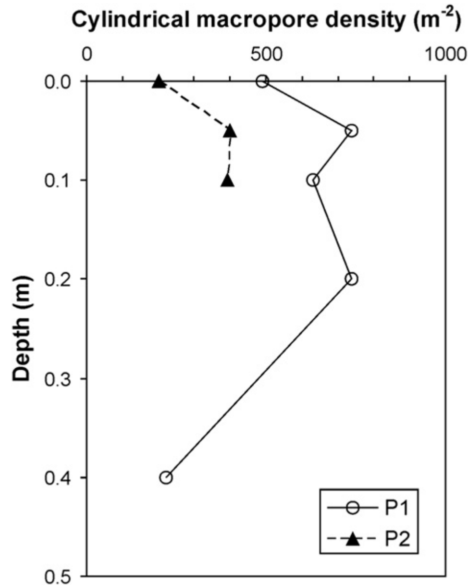

Fig. 3 – Number of cylindrical macropores >2 mm in diameter at infiltration sites P1 and P2, enumerated by depth.

P2 = 0.21 m^3 m^{-3}). The TDR probes were placed in very close proximity; however, small-scale variations in soil structure resulted in slightly different recorded values for soil moisture content. The advance of the non-uniform wetting front resulted in heterogeneous wetting of the soil. During both infiltration events, measured infiltration rates and soil water contents increased substantially when the second pressure head (Ψ_2) was applied, which is consistent with increased soil saturation and the onset of flow in larger soil pores at small negative pressures (−0.8 cm). The infiltration experiment at P2 took three times longer to complete, despite being conducted at initial soil water contents and applied pressures similar to those at P1. The mean infiltration rate and subsequent maximum depth of dye penetration at plot P1 were greater than at P2. These differences in flow and transport can be largely attributed to differences in soil structure, resulting in a greater degree of preferential flow at location P1 compared with P2.

3.4. *Dye distribution in soil after infiltration*

The distribution of dye in soil delineated the zone of infiltration beneath the TI and marked preferential flow channels in the soil profile. Photographs of the central vertical faces (P1–V0, P2–V0) are shown in Fig. 5 and horizontal faces excavated at 5 cm depth (P1–H05, P2–H05) are shown in Fig. 6. These photographs of exposed soil planes have been enhanced by overlaying the binary image analysis of dyed vs. non-dyed soil to make the dye stained areas more visible. The observed dye staining at P1 and P2 showed distinct differences that corresponded with differences in soil macroporosity. At P1 the transport solution penetrated deeper into the subsurface with less lateral infiltration than at P2, despite identical conditions of application. At both locations preferential flow paths were active in transport, but not all of the available macropores were hydrologically active. Generally, dye staining at depths greater than 10 cm was observed to be primarily associated with the presence of cylindrical macropores greater than 2 mm in diameter. The macropores consisted mainly of earthworm burrows and root channels, many of which were observed to be coated with organic-rich material, similar to those observed in Cey et al. (2009). In the top 20 cm of the soil profile, the average density of cylindrical macropores (>2 mm) in plot P1 (650 macropores m^{-2}) was nearly double that in plot P2 (330 macropores m^{-2}). Although the dye patterns suggest transport was dominated by macropore flow at both infiltration sites, the greater macropore density at P1 appeared to contribute to the larger infiltration rates and greater depth of dye staining at this location. The excavation at P1 continued to 60 cm depth, but the dye ended at 58.5 cm at a surface of a large rock. The two earthworm burrows that transported dye to the deepest points at this location were both found to have mature earthworms occupying them near the bottom of the excavation, indicating that these macropores were continuous from the soil surface or close to it.

3.5. *Transport of colloids in soil*

Soil samples collected during excavations were analysed for the presence of all applied colloids to characterize their spatial distribution in soil. Sampling locations for the excavated faces presented were mapped on the images (Figs. 5 and 6) along with the corresponding concentrations of *E. coli* RS2g and MS-1 in soil normalized to the total applied. Negative control samples did not contain any colloidal tracers, indicating that cross-contamination during sampling was not an issue. As described above, the initial number of *E. coli* RS2g bacteria applied to the soil surface was about two orders of magnitude greater than each type of microsphere. For this reason the number of cells detected in each sample tended to be greater than the number of microspheres, but when normalized to the total applied the concentrations detected were very

Table 3 – Characteristics of the controlled infiltration events and depth of dye staining.

ID	Pressure head (Ψ) (cm)	Time interval of infiltration (min)	Mean infiltration rate (cm^3 min^{-1})	Mean initial soil water content (θ)	Mean max soil water content (θ)	Max depth of dye (cm)
P1	−20.4	22	0.84	0.22	0.42	58.5
	−0.8	24	38.6			
P2	−18.5	24	3.8	0.21	0.36	18
	−0.8	116	7.6			

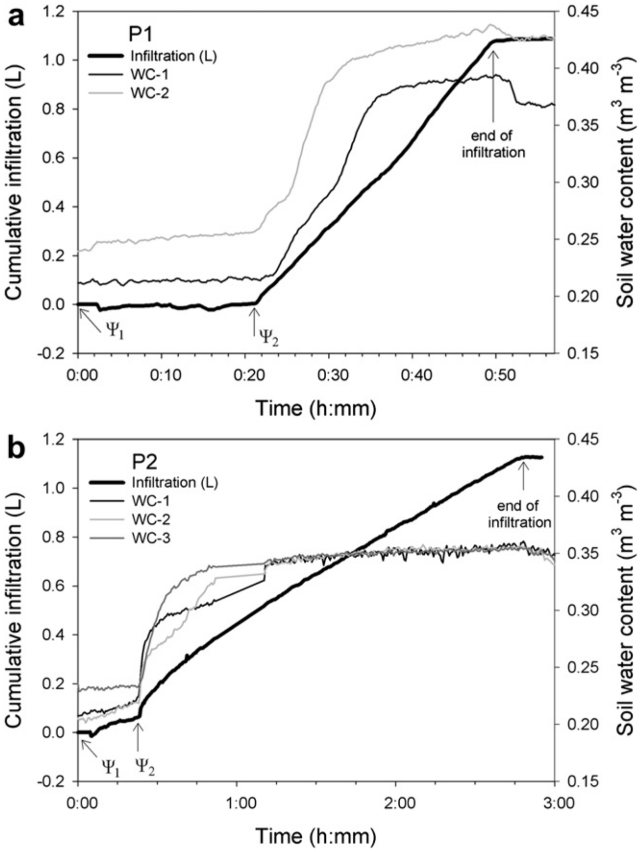

Fig. 4 – Cumulative infiltration curve and changes in soil water content (WC) during infiltration at (a) P1 and (b) P2. Changes in applied tension are indicated on the graph. Note the difference in time scale between graphs.

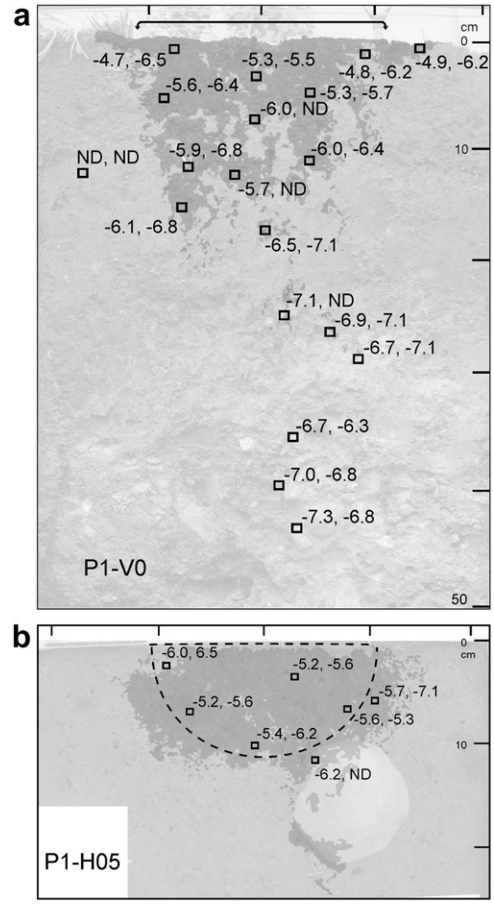

Fig. 5 – Excavated soil faces of (a) P1–V0 and (b) P1–H05 with sampling locations and concentrations of *E. coli* RS2g and MS-1, respectively, normalized to the total applied of each and log transformed. The infiltration area is shown by a black line and arrows or a dashed outline (ND = no detection).

similar. Classic colloid filtration theory (Yao et al., 1971) suggests that the extent of colloid removal is independent of the suspended colloid concentration. Although several laboratory studies have reported concentration dependent removal rates for colloids transported in granular porous media (Bai et al., 1997) and decreases in colloid retention with increasing bulk colloid concentrations, much of this behaviour has been reported when colloid concentrations were high enough to potentially result in blocking ($>\sim 10^7$ (bio)colloids/mL). This is a condition not likely to be relevant in the present investigations; accordingly, the data obtained during the present investigation were evaluated assuming that the retention and transport of the colloids investigated was not affected by their initial applied concentration.

The normalized concentrations of the tracer colloids at various depths in the soil are presented in Fig. 7. As would be expected of a fine-grained medium, a large portion of the colloids were attenuated in the top few centimetres of soil. In both P1 and P2, >2 log reduction of the colloids from the soil surface to 5 cm depth was observed. As transport continued past that depth, the concentrations of colloids remained relatively constant in the dye stained soil, which was primarily associated with preferential flow channels. Natsch et al. (1996) and Burkhardt et al. (2008) reported similar results from colloid transport experiments conducted in the field in agricultural soil using Brilliant Blue dye. Natsch et al. (1996) demonstrated a ∼2 log reduction in *P. fluorescens* concentrations in soil from the surface to 0.2 m depth in two different plots. The colloid and bromide tracer were detected up to 1.5 m depth. Using the ratio of these two tracers, they found that only in the top 0.2 m were the bacteria retained to a greater degree than the conservative tracer. Along macropores from 0.2 to 1.5 m depth there was little variation in the *P. fluorescens*/bromide ratio, indicating that in macropores the retention of both tracers was similar. Burkhardt et al. (2008) recorded ∼1 log reduction in microsphere concentrations in the top 0.2 m of soil, and after a second infiltration microspheres demonstrated consistent concentrations in macropores from 0.5 to 1.5 m depth. There is considerable evidence for a constant retention rate of colloids as they move through macropores.

P1 and P2 had similar concentrations of *E. coli* RS2g, MS-1 and MS-2 in dyed soil from the soil surface to 20 cm depth (Fig. 7); P1 had greater vertical extent of transport. Samples collected below the maximum extent of visible dye on both vertical faces were positive for *E. coli* RS2g and MS-1, indicating that these colloids could be transported with the infiltrating

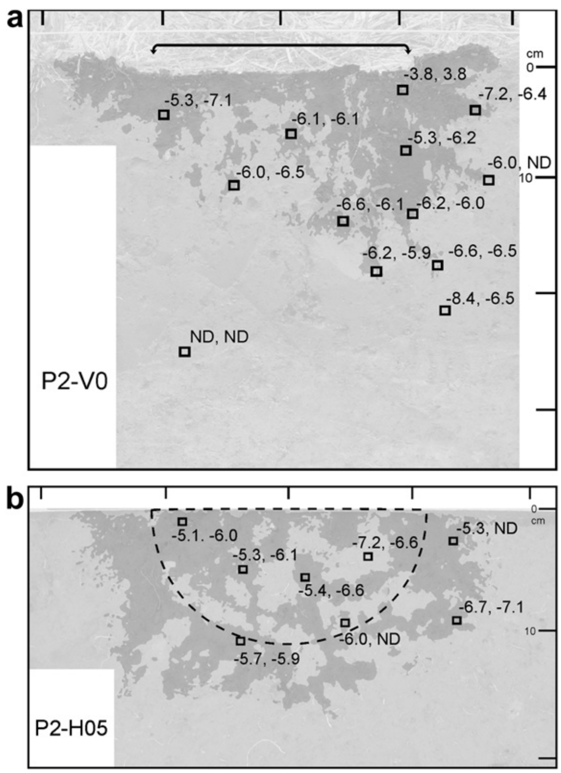

Fig. 6 – Excavated soil faces of (a) P2–V0 and (b) P2–H05 with sampling locations and concentrations of *E. coli* RS2g and MS-1 respectively, normalized to the total applied of each and log transformed. The infiltration area is shown by a black line and arrows or a dashed outline (ND = no detection).

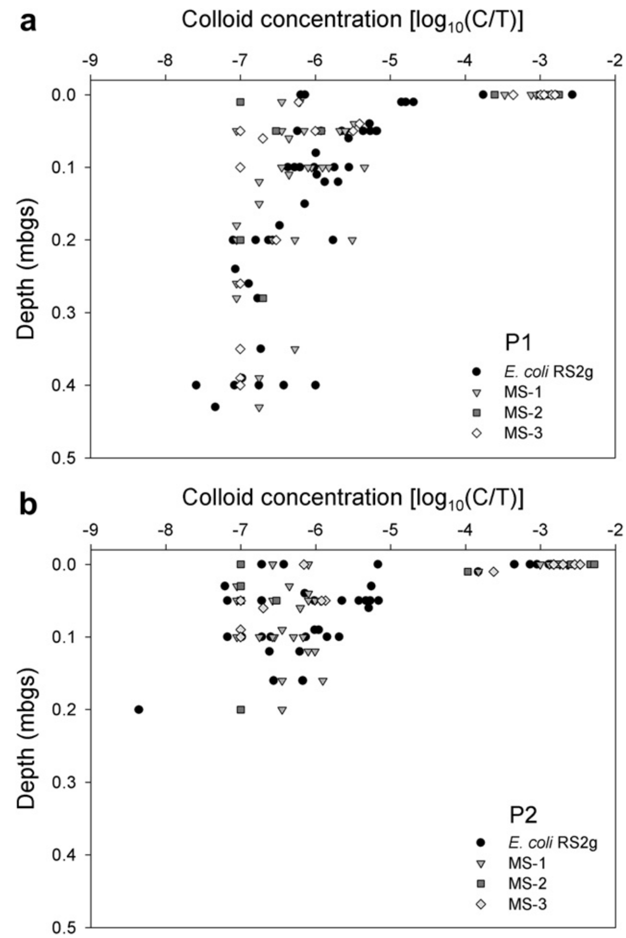

Fig. 7 – Concentration of colloids in soil samples by depth (meters below ground surface) at (a) P1 and (b) P2, normalized to the total amount applied of each (counts per g soil/total applied) and log transformed.

water farther than the dye; however, the dye can adsorb to soil surfaces and is not a completely conservative tracer for pore water movement. *E. coli* RS2g appeared to be the most mobile colloid, detected at greater distances from the surface application than any of the microspheres; however, significantly more cells were added and this may have made RS2g easier to detect in areas of low concentration. MS-1 was the microsphere with the greatest mobility and most closely matched the distribution of *E. coli* RS2g in the subsurface: a result that would be expected given that, of the microspheres studied, MS-1 is the closest in size to the *E. coli* RS2g; moreover, its zeta potential is similar to that of the *E. coli* RS2g. MS-2 was transported the least, despite it not being the largest microsphere; however, its relatively low zeta potential likely contributed to this outcome because it resulted in MS-2 experiencing the least electrostatic repulsion when approaching negatively charged soil surfaces and organic matter.

The distribution of colloids in soil can be described in relation to macropores and dye staining. Macropores conducted solution and colloids to depth, and their large size (>2 mm) permitted the potential passage of all applied colloids. Colloid presence was positively related with the intensity of dye staining (Table 4). At site P1 on excavated face P1–H05 (Fig. 5b) each colloid type was detected in the most heavily dyed areas and large dyed earthworm burrows. The lightly dyed areas at this depth, associated mainly with the soil matrix, contained only RS2g and MS-1. At 10 cm depth and below, MS-2 and MS-3 were detected only in relation to dye stained macropores, although not necessarily at all locations. The sample in dye category 1 and positive for MS-2 was a lightly dyed soil sample collected from the 10 cm depth horizontal slice of P1, where roots had penetrated a fracture and been dyed blue by the infiltrating solution. Every dyed sample from P1–H20 contained *E. coli* RS2g, and most of these also contained MS-1. The deepest detection of MS-2 at P1 was

Table 4 – Number of soil samples positive for each colloid, as ranked by categories of dye staining (2: ≥0.05 mg dye g^{-1} soil; 1: <0.05 and ≥0.00005 mg dye g^{-1} soil; 0: no visible dye in sample; n = 84).

Dye	n[a]	RS2g	MS-1	MS-2	MS-3
2	36	36	30	19	24
1	37	37	28	1	6
0	11	11	7	2	2

a n = number of samples in the dye classification category.

28 cm; below this depth most samples contained only *E. coli* RS2g and MS-1, with occasional detections of MS-3 in earthworm burrows. Each of the two samples positive for MS-2 in dye category 0 (i.e. no visible dye) had been sampled from deeper locations in P1 and P2, below the bulk of the dye stained soil. Two samples that were positive for MS-3 in dye category 0 were the second and third deepest samples collected from face P1–V0. These samples were associated with macropore flow channels and indicate the ability of a relatively small amount of transport solution to mobilize colloids to depth.

It was possible to follow patterns of colloid distribution in larger burrows between slices, as these macropores were relatively linear features. One particular burrow at P1 was observed at P1–H10, but not at shallower depths in the soil profile. Flow into it was probably initiated by a heavily dye stained mass of roots observed directly above it at P1–H05, in which all colloids were detected. In the burrow at P1–H10 the concentrations of all detected colloids were reduced and no MS-2 was present. At 20 cm depth, only *E. coli* RS2g cells and MS-1 were detected, and at 40 cm depth MS-3 and *E. coli* RS2g were present, but without MS-1. These potential inconsistencies in colloid detection at depth may be related to the low concentrations present, and may reflect the limits of detection more than absolute trends in distribution.

Fewer macropores were available for colloid transport at P2. An earthworm burrow visible at P2–H05 had all colloid types present in it at that depth. The burrow extended to P2–H10, but by 10 cm depth MS-2 and MS-3 were absent and only *E. coli* RS2g and MS-1 remained. Other heavily dyed areas tended to have only *E. coli* RS2g and MS-1, while lightly dyed soil usually only had RS2g cells present.

To quantify the ability of microspheres to act as a surrogate for *E. coli* RS2g, samples positive for the bacteria were related to the presence of microspheres ($n = 84$). The MS-1 microspheres were present in 77% of the samples that tested positive for *E. coli* RS2g. The poorest predictor of *E. coli* RS2g transport was MS-2, as it was present in only 26% of the samples in which RS2g cells were found. MS-3 was detected 38.1% of the time *E. coli* RS2g was found. Clearly MS-1 was the best ''predictor'' of *E. coli* RS2g cell presence and transport during the present investigation.

Relative to laboratory comparisons, very few field studies have directly compared the transport of microorganisms with microspheres. Harvey et al. (1993) compared the transport of 0.74 μm microspheres with indigenous bacteria in a sandy aquifer and had mixed results depending on the depth of injection. For two of three depths microorganisms and microsphere transport was very similar, but at one depth there was greater retention of microspheres. This variability was attributed to physical heterogeneity of the aquifer. In the same aquifer Harvey et al. (1995) found 2.0 μm microspheres were useful analogs for the transport of flagellated protozoa in the field and this was confirmed by laboratory experiments. Sinreich et al. (2009) explored vadose zone transport from the soil surface through 10 m of limestone and generated breakthrough curves for 1.0 μm microspheres and motile groundwater bacteria that were similar to each other; however, more microspheres were retained relative to the initial applied concentrations and there was also much less mass recovery of microspheres, indicating that the bacteria were more mobile in the vadose zone. These studies, while conducted under different conditions and sampled using different techniques, demonstrate the potential for the use of microspheres in the investigation of transport pathways and various aspects of abiotic transport behaviour, but only when they are well-characterized and imitate closely the properties of the colloid in question. As the scale of inquiry increases in time and space, differences in microsphere and microorganism properties may become more apparent (Harvey et al., 1995; Sinreich et al., 2009).

4. Conclusions

A protocol was successfully developed for the application, recovery and analysis of microspheres and *E. coli* RS2g bacteria in soil samples. The methods used were successful in generating concentration data over the range of five orders of magnitude. These experimental procedures may be tailored to different types of colloids and porous media. Survival and irreversible attachment should be considered when conducting colloid transport experiments because recovery can be affected by both these factors.

E. coli RS2g and microsphere transport in this field experiment occurred under conditions of unsaturated flow at two field sites with distinctly different soil characteristics. Despite differences in infiltration rates, soil properties, dye patterns, and depth of movement, concentrations of colloids in dye stained soil were similar at both locations. Colloid properties and macropore density were the most important factors affecting colloid transport. Preferential transport along macropores was important in transporting dye and colloids to depth. Mechanisms of retention were very effective in the near-surface, where colloid concentrations in dyed soil decreased by more than two orders of magnitude. Once past 5 cm depth, total transport mass diminished with depth, but colloid concentrations in areas of dyed soil remained fairly consistent with depth. The colloids were attenuated in soil in the order of MS-2 > MS-3 > MS-1 > *E. coli* RS2g. The movement of colloids in the subsurface is highly complex, and it may be difficult to find any one microorganism or colloid to reliably predict the transport of any other in a given medium. However, when selected based on similarity of size and surface properties, microspheres can be a useful surrogate for bacteria to explore and delineate potential pathways of transport.

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